Oluwatosin Ayeni

Member of Technical Staff: Inference Research — Doubleword

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PROFILE

AI Inference Research Engineer with production experience optimising LLM serving across vLLM, SGLang, and TensorRT-LLM on H100 and RTX 4090 hardware. Benchmarked 1,443× TTFT speedup via RadixAttention; profiled CUDA KV cache kernels for 3.8× throughput gain; ran end-to-end speculative decoding and quantisation experiments. Interested in multi-LoRA serving, memory-constrained GPU inference, and the research-to-production pipeline. MSc AI with Distinction, Strathclyde 2023.

CORE COMPETENCIES

Inference Engines	vLLM, SGLang, TensorRT-LLM, ONNX Runtime, Hugging Face Transformers/TGI
GPU & CUDA	NVIDIA Nsight Systems/Compute, CUDA kernel profiling, memory bandwidth analysis, kernel fusion, warp occupancy, BFloat16/FP8
Inference Research	Speculative decoding, RadixAttention/prefix caching, AWQ/GPTQ quantisation, KV cache compression, multi-LoRA serving, flash attention, continuous batching
LLM Fine-tuning	PyTorch, PEFT/LoRA, Mistral 7B, Qwen3, Phi-4, Gemma 2 — full pipeline: data prep, training, eval, inference
Infrastructure	Kubernetes, Docker, FastAPI, Prometheus, Grafana, Python, GitHub Actions

PROFESSIONAL EXPERIENCE

Jan 2025–present **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-benchmark (H100) vLLM vs SGLang, H100 NVL, Qwen3-8B BF16 TP=2. RadixAttention: 1,443× TTFT speedup. Peak: 5,965 tok/s. Live results

open-model-engine Inference benchmarking: vLLM, SGLang, TensorRT-LLM — CUDA profiling, quantisation, speculative decoding. GitHub

inference-harness Unified benchmark harness: closed + open serving engines, GPU profiling, automated dashboard. GitHub

lab-vllm-omni vLLM-Omni phased inference lab: all 5 output modalities — text, audio, image, video, structured output. GitHub

My_finetuned_Phi4 Phi-4 LoRA fine-tuning: full pipeline — data prep, training, eval, inference. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale